WEEK 7 / LESSON 6 / 50 MINUTES

L6

Mobile · IoT · Jetson

Edge AI is where latency, privacy, and cost converge. When your Hindi voice assistant must work in a mobile-network dead zone — edge is the only answer.

When did YOUR phone last use AI completely offline?

WHAT'S INSIDE

WHY THIS LESSON MATTERS

JioBharat 5G — 1 million devices running AI offline in a train tunnel.

WHY
Real
A real reason this matters in industry today.

SCALE
Today
A real reason this matters in industry today.

SPEED
Now
A real reason this matters in industry today.

IMPACT
Yours
A real reason this matters in industry today.

LEARNING OUTCOMES

By the end of this lesson, you will be able to —

1. When edge beats cloud.
2. Convert PyTorch → ONNX.
3. Quantise to int8 / fp16.
4. Deploy on mobile + Jetson.
5. Measure on-device latency.

AGENDA

What we'll cover today

Short, sequenced parts. Each one builds on the last.

01
Three drivers to pick edge

Key concept of this lesson.

02
ONNX — the portable format

One format; all devices

03
Quantisation levels

A 7B LLM: fp16 = 14 GB. int4 = 3.5 GB. Same model runs on a phone.

04
Deploy targets

Key concept of this lesson.

01

PART 1 OF 4

Three drivers to pick edge

Key concept of this lesson.

CONCEPT 1 OF 4

Three drivers to pick edge

A core idea you'll use throughout this lesson.

DEFINITION

Three drivers to pick edge

EXAMPLE

Real-world example: applies directly to Indian industry contexts.

CONCEPT 2 OF 4

ONNX — the portable format

torch.onnx.export saves any PyTorch model.
Load with ONNX Runtime on ANY device.
iOS, Android, Jetson, browser — same file works.
Dynamic axes for variable batch / sequence.
PORTABILITY
ONNX

Train once; run anywhere.

KEY POINTS

- torch.onnx.export saves any PyTorch model
- Load with ONNX Runtime on ANY device
- iOS, Android, Jetson, browser — same file works
- Dynamic axes for variable batch / sequence

CONCEPT 3 OF 4

Quantisation levels

A 7B LLM: fp16 = 14 GB. int4 = 3.5 GB. Same model runs on a phone.

DEFINITION

Quantisation levels

A 7B LLM: fp16 = 14 GB. int4 = 3.5 GB. Same model runs on a phone.

EXAMPLE

Real-world example: applies directly to Indian industry contexts.

CONCEPT 4 OF 4

Deploy targets

A core idea you'll use throughout this lesson.

KEY POINTS

- A core idea you'll use throughout this lesson
- Practical, named, applied to a real Indian use-case.
- Practical, named, applied to a real Indian use-case.

02

PART 2 OF 4

Engage and reflect

Pause, talk, and connect what you just learned to your own work.

✦ THINK · PAIR · SHARE

ACTIVITY

fp32 vs int8 on one model

PROMPT

Apply the four concepts above to one real example from your own week.

HOW TO RUN IT

1.
2.
3.
4. Reveal: instructor confirms the canonical answer.

REFLECT & APPLY

Three questions before you close this lesson

Spend 60 seconds on each. Write the answers down — no scrolling, no shortcuts.

Q1 *Which app in your life is cloud-only that could be offline?*

Q2 *Where does thermal throttling silently bite you?*

Q3 *Would a Jetson Nano solve a problem in your work?*

03

PART 3 OF 4

Knowledge check

Three short questions. One minute each. Honest answers only.

KNOWLEDGE CHECK · QUESTION 1 OF 3

Three drivers for edge:

- **A** A,B,C
- **B** Latency, privacy, cost
- **C** Speed, cost, size
- **D** Memory, GPU, CPU

ANSWER · QUESTION 1 OF 3

Three drivers for edge:

CORRECT ANSWER

B

Latency, privacy, cost

WHY

These three together justify edge over cloud

KNOWLEDGE CHECK · QUESTION 2 OF 3

ONNX purpose:

A Training

B Portability across frameworks/devices

C Compression

D UI

ANSWER · QUESTION 2 OF 3

ONNX purpose:

CORRECT ANSWER

B

Portability across frameworks/devices

WHY

Train anywhere; run anywhere

KNOWLEDGE CHECK · QUESTION 3 OF 3

int8 vs fp32 memory:

A Same

B 2×

C 4×

D 8×

ANSWER · QUESTION 3 OF 3

int8 vs fp32 memory:

C

CORRECT ANSWER

4×

WHY

25% of fp32

04

PART 4 OF 4

Apply and close

One thing to remember. One thing to ship this week.

SUMMARY

Five sentences to remember

If you remember nothing else from this lesson, remember these five lines.

1. Pick the right tool — never the loudest one.
2. Practice the framework on a real example.
3. Watch for the three classic pitfalls.
4. Document your decision in one sentence.
5. Carry the discipline forward to the next lesson.

WORKED EXAMPLE

How this lesson plays out in one real Indian context

THE SCENARIO

A team applies the lesson's main idea to a real workflow — and the results show up within a quarter.

THE TAKEAWAY

"Laptop benchmarks lie."

WHAT TO NOTICE

1. Three drivers to pick edge
2. ONNX — the portable format
3. Quantisation levels
4. Deploy targets

APPLY IT → ASSIGNMENT

Quantise + measure on Pi

W3 model to Raspberry Pi 4.

DELIVERABLES

- Export ONNX.
- Quantise int8.
- Measure latency table.
- Submit.

WHAT 'GOOD' LOOKS LIKE

Deliverable: Latency table on forum.

ONE THING TO REMEMBER

"Laptop benchmarks lie."

— AI Learning Hub

END OF LESSON 7

What you can now do

Each one of these is now part of your working vocabulary.

1. Pick the right tool — never the loudest one.
2. Practice the framework on a real example.
3. Watch for the three classic pitfalls.
4. Document your decision in one sentence.
5. Carry the discipline forward to the next lesson.

NEXT → Lesson 7 — Indic NLP.